

POSTS AND TELECOMMUNICATIONS INSTITUTION OF TECHNOLOGY

FACULTY OF INFORMATION TECHNOLOGY I

COURSE: INFORMATION SYSTEM ANALYSIS AND DESIGN



ASSIGNMENT 01

REPORT OF

E-COMMERCE INFORMATION SYSTEM INVESTIGATION

Student name: NGUYỄN MẠNH HÙNG

Student code: B23DCCE040

Instructor: TRẦN ĐÌNH QUẾ

Hà Nội - 2026

1. Introduction

Information systems play an essential role in supporting modern e-commerce businesses by managing business activities, information, and stakeholder interactions efficiently. However, effective system development should begin with understanding the real-world business problem rather than focusing immediately on software design or implementation.

This report presents the investigation conducted in Assignment 01, focusing on a realistic e-commerce business. It examines the business context, key activities, stakeholders, information needs, and problems in current processes. Based on these findings, the report defines the preliminary scope, boundaries, and objectives of the proposed information system.

The purpose of this investigation is to identify the problems the system should address and how it can better support the organization and its stakeholders. The findings provide a foundation for Assignment 02, where these business problems and stakeholder needs will be translated into formal system requirements.

2. Business Background

2.1. Organization

JULY Online Shop is a medium-sized B2C e-commerce business that sells a variety of beauty products directly to individual customers through digital online channels. The primary business purpose is to provide customers with a convenient and reliable environment to discover, purchase, and receive beauty products online. Its main sales channel is online shopping platform, where customers can browse products, manage shopping carts, place orders, make payments, and track deliveries.

To support its business operations, JULY Online Shop manages major activities including product management, customer management, order processing, payment management, inventory management, shipping management, review management, promotion management, and sales reporting. The company's main business objectives are to provide convenient online purchasing, maintain accurate product availability, process orders efficiently, ensure reliable delivery, and achieve customer satisfaction while generating revenue from product sales.

2.2. Business Model

Organization Name: JULY Online Shop.

Type of Business: JULY Online Shop is a medium-sized B2C (Business-to-Consumer) e-commerce business that sells beauty products directly to individual customers through online channels.

2.3. Products and Customers

Products: JULY Online Shop offers a variety of product categories: perfumes, makeup, skincare, and haircare.

Customers: The primary target customers are individual consumers who purchase beauty products online. Customers can browse product information, add products to their shopping carts, place orders, make payments, and receive products through delivery services.

2.4. Major Business Activities

The company relies on several major business activities to maintain its daily operations and serve its stakeholders:

- **Product management:** Creating and maintaining accurate product information, including product names, descriptions, prices, images, and product categories, so that customers can make informed purchasing decisions.
- **Customer Management:** Managing customer accounts, profiles, contact information, delivery addresses, and order history, providing support related to products and orders.
- **Review management:** Collecting, storing, and managing customer ratings and reviews after purchases to evaluate customer satisfaction and provide feedback for improving products and services.
- **Order Management:** Managing the products selected by customers in their shopping carts and processing placed orders, including order creation, confirmation, status updates, and cancellation when applicable.
- **Payment Management:** Processing customer payments through external payment providers, recording transaction information, and updating payment statuses to confirm whether orders have been successfully paid.
- **Inventory Management:** Tracking stock levels and updating product availability based on product sales and order processing to ensure that customers can only purchase available products.
- **Shipping Management:** Coordinating the dispatch of confirmed orders with external shipping providers, tracking shipments, and updating delivery statuses until orders are delivered to customers.
- **Promotion Management:** Creating and managing promotional offers, discount codes, and eligibility conditions to attract customers and encourage product purchases.
- **Reporting:** Consolidating information from sales, orders, payments, inventory, customers, and other business activities into automated reports to monitor business performance and support management decision-making.

3. Stakeholder Analysis

3.1. Internal Stakeholders

Internal stakeholders are individuals who work within JULY Online Shop and are responsible for carrying out its daily business activities. The primary internal stakeholders include:

- **Administrator:** Manages product, customer, order, promotion, and system information.

- **Sales Staff:** Handles customer inquiries and supports order processing and customer-related activities.
- **Warehouse Staff:** Manages inventory, prepares products for dispatch, and updates stock information.
- **Store manager:** Oversees business performance and uses operational information and reports to support decision-making.

3.2. External Stakeholders

External stakeholders are individuals or organizations outside JULY Online Shop that interact with the business or support its operations. The primary external stakeholders include:

- **Customers:** Browse and purchase beauty products and provide feedback about their purchases.
- **Suppliers:** Provide beauty products and supply information required to maintain inventory.
- **Payment Providers:** Process customer payments and provide transaction results.
- **Shipping Providers:** Collect, transport, and deliver customer orders and provide delivery status information.

3.3. Stakeholder Interests

Understanding the roles and primary needs of each stakeholder is critical for defining the scope and objectives of the proposed information system. The table below summarizes the main stakeholders and their interests:

Stakeholder	Role	Main Interest / Need
Customer	Browses and purchases beauty products	Convenient shopping, accurate product information, secure payment, and reliable delivery
Administrator	Manages business and system information	Efficient and accurate management of products, customers, orders, and promotions
Sales Staff	Handles customer inquiries and supports orders	Accurate customer, product, and order information to provide timely customer support

Warehouse Staff	Manages inventory and prepares orders for dispatch	Accurate, real-time stock information and clear order details
Store manager	Monitors business	Accurate reports on sales, orders, inventory, and customer activity
Payment Provider	Processes payments	Secure and accurate payment and transaction information
Shipping Provider	Collects and delivers customer orders	Accurate delivery addresses, shipment details, and order information
Supplier	Provides beauty products and supply information	Accurate product, order, and inventory requirements

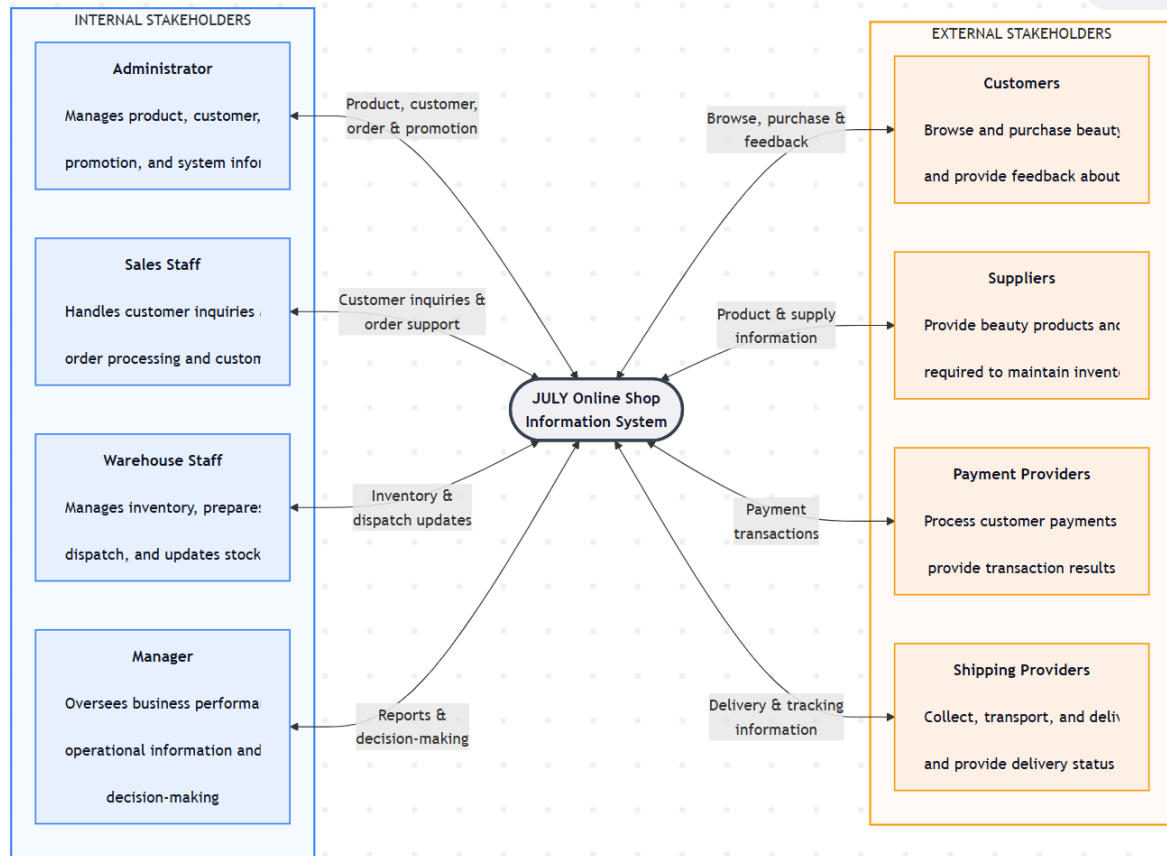


Figure 3.1: Stakeholders map

4. Business Activity Analysis

4.1. Product Management

Staff identifies and collects information for new products → Staff enters or updates product information → Information is reviewed for accuracy and completeness → Products are made available for customers to browse → Information is updated when product details or availability change.

4.2. Customer Management

Customer creates an account or provides information during checkout → Customer information is recorded and maintained → Information is used for order processing and delivery → Order history is associated with the customer's account → Staff accesses customer information to handle inquiries and order issues.

4.3. Order Management

Customer browses available products → Shop displays product details and availability → Customer compares and selects products → Customer checks availability and applicable promotions → Customer adds the selected product to the shopping cart.

4.4. Payment Management

Customer selects a payment method at checkout → Payment information is sent to the payment provider → Provider processes the transaction and returns the result → Shop records the transaction and payment status → Order proceeds if payment is successful; failed or pending payments are handled accordingly.

4.5. Inventory Management

Stock levels are recorded for each product or variant → Product availability is provided to customers → Required stock is identified when an order is confirmed → Stock is reduced or reserved and the updated level is recorded → Low or out-of-stock products are marked accordingly → Inventory information supports future orders and reporting.

4.6. Shipping Management

Confirmed orders are prepared for dispatch → Order and delivery information is sent to the shipping provider → Package is collected and tracking information is recorded → Shipment is transported to the customer's address → Delivery status is updated until the product is received and the order is marked as delivered.

4.7. Reporting

Business activities generate operational data → Relevant data is collected, consolidated, and analyzed → Reports are generated and provided to store managers or relevant staff → Management reviews reports to monitor performance and support decision-making.

4.8. Promotion Management

Staff identifies suitable promotional opportunities → Promotion details and conditions are created → Promotion is made available to customers → Customer applies a promotion during shopping or checkout → System checks the conditions and applies the applicable discount.

4.9. Review Management

Customer receives the purchased product and submits a rating or review → Review is recorded and associated with the customer and product → Business checks and moderates the review when necessary → Approved reviews are stored and displayed to customers → Staff

and management analyze feedback to improve products, services, and operations.

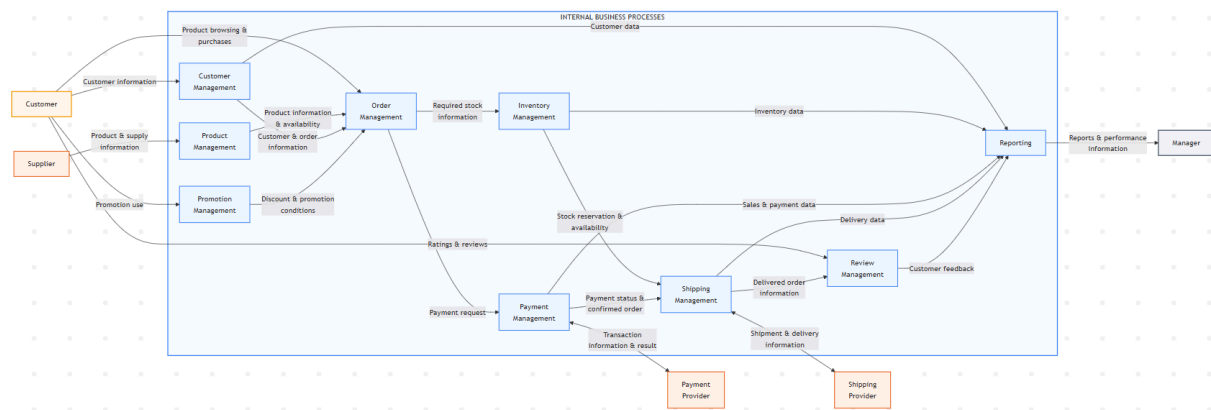


Figure 4.1: Diagram of High-Level business process

5. Information Analysis

5.1. Information Produced

- **Product Information:** Product details, prices, images, categories, variants, and availability.
- **Customer Information:** Customer profiles, contact details, addresses, and order history.
- **Shopping Cart Information:** Selected products and quantities.
- **Order Information:** Order details, products, quantities, value, delivery address, and status.
- **Payment Information:** Payment transaction details and status.
- **Inventory Information:** Stock quantities, status, and availability changes recorded.
- **Shipment Information:** Shipment details, tracking numbers, and delivery status.
- **Review Information:** Ratings, reviews, and product feedback.
- **Promotion Information:** Promotional offers, discount codes, and eligibility conditions.
- **Sales Information:** Sales totals, order volumes, and sales-related data generated from completed transactions.
- **Customer Support Information:** Customer inquiries, responses, support requests, and resolution details
- **Business Reports:** Consolidated business information and performance reports.

5.2. Information Consumed

- **Product Information:** Used for browsing products, support customers, manage stock, and fulfill orders.
- **Customer Information:** Used for manage customer accounts, handle inquiries, and support customer-related activities.
- **Shopping Cart Information:** Used to review and modify selected products before checkout.

- **Order Information:** Used to process orders, prepare products, process payments, and fulfill deliveries.
- **Payment Information:** Used to confirm payment status and support order processing.
- **Inventory Information:** Used to determine product availability, manage stock, and support order fulfillment.
- **Shipment Information:** Used to track shipments, monitor delivery progress, and confirm order delivery.
- **Review Information:** Used to evaluate customer satisfaction, respond to feedback, and identify areas for improvement.
- **Promotion Information:** Used to apply and manage eligible promotional offers and discounts.
- **Sales Information:** Used to monitor sales activities and evaluate business performance.
- **Business Reports:** Used to monitor business performance and support operational decision-making.
- **Customer Support Information:** Used to review customer inquiries, provide support, and track issue resolution.

5.3. Information Stored

- **Product Information:** Stored to maintain the product catalogue and provide consistent product information to customers.
- **Customer Information:** Stored to maintain customer accounts, delivery details, and order history.
- **Shopping Cart Information:** Stored temporarily or persistently to maintain customers' selected products before checkout.
- **Order Information:** Stored to track orders, maintain order history, and support order fulfillment.
- **Payment Information:** Stored to maintain transaction records and payment status associated with orders.
- **Inventory Information:** Stored to monitor stock quantities and product availability.
- **Shipment Information:** Stored to track shipments, current status of orders and maintain delivery records.
- **Review Information:** Stored to maintain customer ratings and reviews associated with products.
- **Promotion Information:** Stored to maintain active and historical promotional offers and discount conditions.
- **Sales Information:** Stored to support sales analysis and reporting.
- **Business Reports:** Stored for future reference, performance monitoring, and management decision-making.
- **Customer Support Information:** Stored to maintain records of customer inquiries, support interactions, responses, and issue resolutions for future reference and consistent customer service.

5.4. Information Catalogue

Information	Produced By	Used By	Treatment	Purpose
Product Information	Supplier, Administrator	Customers, Sales Staff, Warehouse Staff, Administrator, Store manager	Exchanged, Stored, Consumed	Provide accurate product details and availability
Customer Information	Customers	Administrator, Store manager, Sales Staff	Exchanged, Stored, Consumed	Manage customer accounts and support customer service
Shopping Cart Information	Customers	Customers	Stored, Consumed	Manage selected products and quantities before checkout
Order Information	Customers	Sales Staff, Warehouse Staff, Payment Provider, Shipping Provider	Exchanged, Stored, Consumed	Support order processing, payment, preparation, and delivery
Payment Information	Payment Provider	Sales Staff, Store manager	Exchanged, Stored, Consumed	Confirm payment status and support order processing
Inventory Information	Warehouse Staff	Customers, Sales Staff, Warehouse Staff, Administrator, Store manager, Supplier	Exchanged, Stored, Consumed	Monitor stock levels and product availability
Shipment Information	Shipping Provider	Customers, Sales Staff, Store manager	Exchanged, Stored, Consumed	Track shipments and monitor delivery progress
Review Information	Customers	Store manager, Sales Staff	Stored, Consumed	Evaluate customer satisfaction and identify improvements

Promotion Information	Administrator, Store manager	Customers, Sales Staff	Stored, Consumed	Apply promotional offers and discounts
Sales Information	Sales staff	Store manager, Customer	Stored, Consumed	Monitor sales activities and business performance
Business Reports	Sales staff	Store manager	Produced, Stored, Consumed	Monitor performance and support decision-making
Customer Support Information	Customers, Sales Staff	Administrator, Store manager	Exchanged, Stored, Consumed	Record inquiries, support requests, responses, and resolutions

6. Current problems:

6.1. Problems identifications:

The five major problems identified at JULY Online Shop are:

1. Inventory information is frequently inaccurate or outdated.
2. Product information is inconsistent across the catalogue.
3. Order status and tracking information are unclear to customers.
4. Customer information is fragmented across different records or locations.
5. Business reports are slow and prone to manual errors.

These problems are closely related because they originate from weaknesses in how business information is recorded, updated, shared, and consolidated.

6.2. Problems Causes:

The main causes of the identified problems are:

- **Inaccurate or outdated inventory information:** Manual and delayed stock updates.
- **Inconsistent product information:** Product details are managed by different staff without a centralized information source.
- **Unclear order status and tracking:** Order and shipment information is handled across separate activities and providers without effective synchronization.
- **Fragmented customer information:** Customer data is stored in separate records rather than a unified system.
- **Slow and error-prone business reports:** Data from different business activities must be manually consolidated.

6.3. Problem Consequences

Problem	Likely Cause	Consequence
Inventory information is inaccurate or outdated	Inventory is updated manually, causing delays between stock changes and available information.	May cause unavailable products to be sold, leading to cancellations and lower customer satisfaction.
Product information is inconsistent	Product details are maintained by different staff without a single consistent source of information.	May confuse customers, reduce confidence, and increase product-related inquiries.
Order status and tracking information are unclear	Order and shipment information is handled across different activities and external providers without effective synchronization.	Makes it difficult for customers to track orders and increases support workload.
Customer information is fragmented	Customer information is stored across different records instead of being managed as a unified set.	Makes customer support less efficient and prevents staff from accessing complete customer information.
Business reports are slow and prone to errors	Information from sales, orders, payments, inventory, and customers must be manually consolidated, increasing processing time and errors.	Delays reliable decision-making about sales, inventory, orders, and promotions.

6.4. Problem Analysis: Problem–Cause–Consequence Relationship

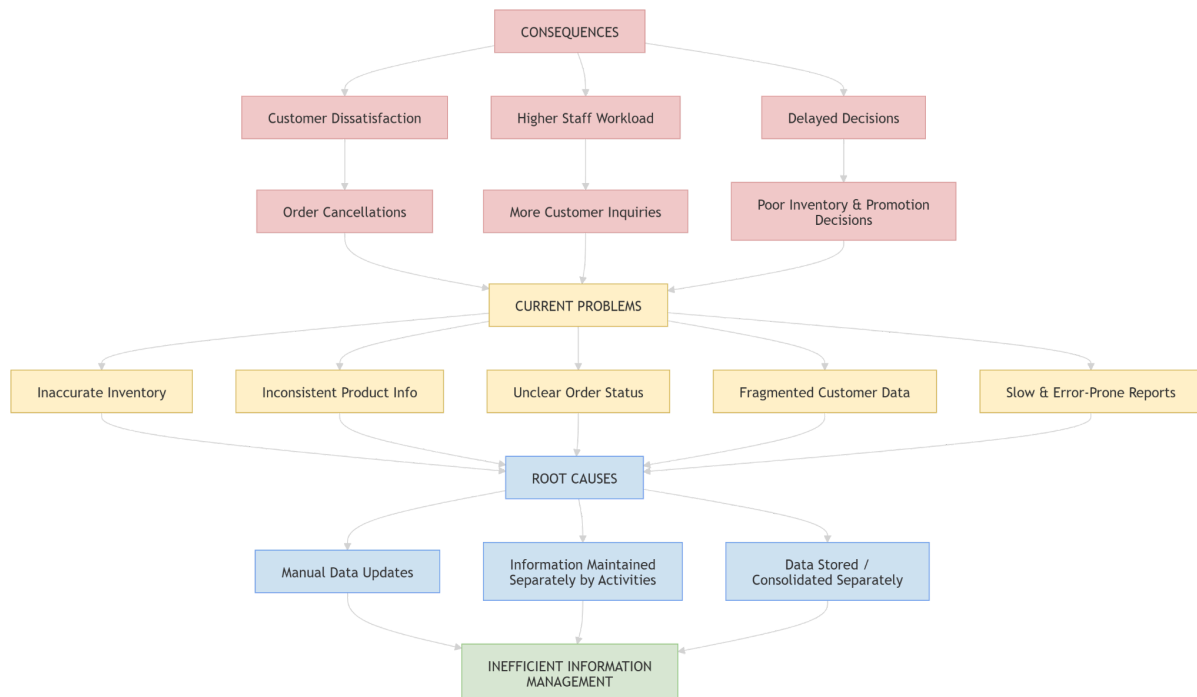
The identified problems are connected through inefficient information management. The main causes, problems, and consequences can be summarized as follows:

- **Customers** may experience inaccurate product availability, unclear order status, and reduced confidence in the shopping process.
- **Sales Staff** may spend additional time responding to customer inquiries and resolving information inconsistencies.
- **Warehouse Staff** may work with inaccurate or outdated stock information.
- **Administrators** may spend additional time maintaining and consolidating business information.

Overall, these problems share a common underlying cause: business information is not centrally and consistently managed across different activities.

6.5. Problem Tree

The following problem tree presents the relationship between the underlying causes, the major information-management problems, and their operational consequences.



7. Proposed Information System

7.1. System Purpose

The proposed E-Commerce Information System aims to centralize and improve business information management at JULY Online Shop by addressing inaccurate inventory, inconsistent product information, unclear order tracking, fragmented customer data, and slow reporting. The system will provide a centralized and consistent source of information, reduce

manual data handling, improve accuracy and timeliness, clarify order information, and support faster reporting and more reliable decision-making.

7.2. System Scope

- **Product & Promotion Management:** Centralizing product information, including names, descriptions, prices, images, categories, variants, and availability, promotion and discount information.
- **Customer & Review Management:** Maintaining customer profiles, contact and delivery information, order history, ratings, and reviews in a unified system.
- **Shopping Cart & Order Management:** Supporting product selection, cart activities, checkout, order creation, confirmation, status updates, and cancellation where applicable.
- **Inventory Management:** Maintaining current stock quantities, product availability, with inventory updates based on confirmed orders and stock changes.
- **Payment Management:** Exchanging payment requests, transaction results with external payment providers, recording payment status in the system. The system will not perform the providers' transaction processing internally.
- **Shipping & Order Tracking Management:** Exchanging order and delivery information with external shipping providers and recording shipment and tracking updates for access by customers and staff.
- **Customer Support Management:** Recording customer inquiries, support requests, staff responses, and issue resolution to support consistent customer-service activities.
- **Reporting:** Consolidating information from sales, orders, payments, inventory, customers, reviews, and other business activities to provide timely reports for management decision-making.

7.3. System Boundary

The system boundary defines the business activities and information managed by the proposed E-Commerce Information System and separates them from external entities and services.

Inside the System Boundary

The proposed system supports:

- Product and promotion management
- Customer and review management
- Shopping cart and order management
- Inventory management
- Payment-status management and payment-provider integration
- Shipping and order-tracking management
- Customer support management
- Sales and business reporting

The system centralizes JULY Online Shop's core business information and supports access by Administrators, Sales Staff, Warehouse Staff, Store manager, and Customers according to their roles.

Outside the System Boundary

- **Payment Providers:** Process payments externally. The system exchanges payment requests and results and records payment status.
- **Shipping Providers:** Handle physical collection, transportation, and delivery. The system exchanges order and delivery information and records tracking updates.
- **Suppliers:** Provide products and product information. The system maintains received information but does not manage suppliers' procurement, production, or inventory operations.

7.4. Context Diagram

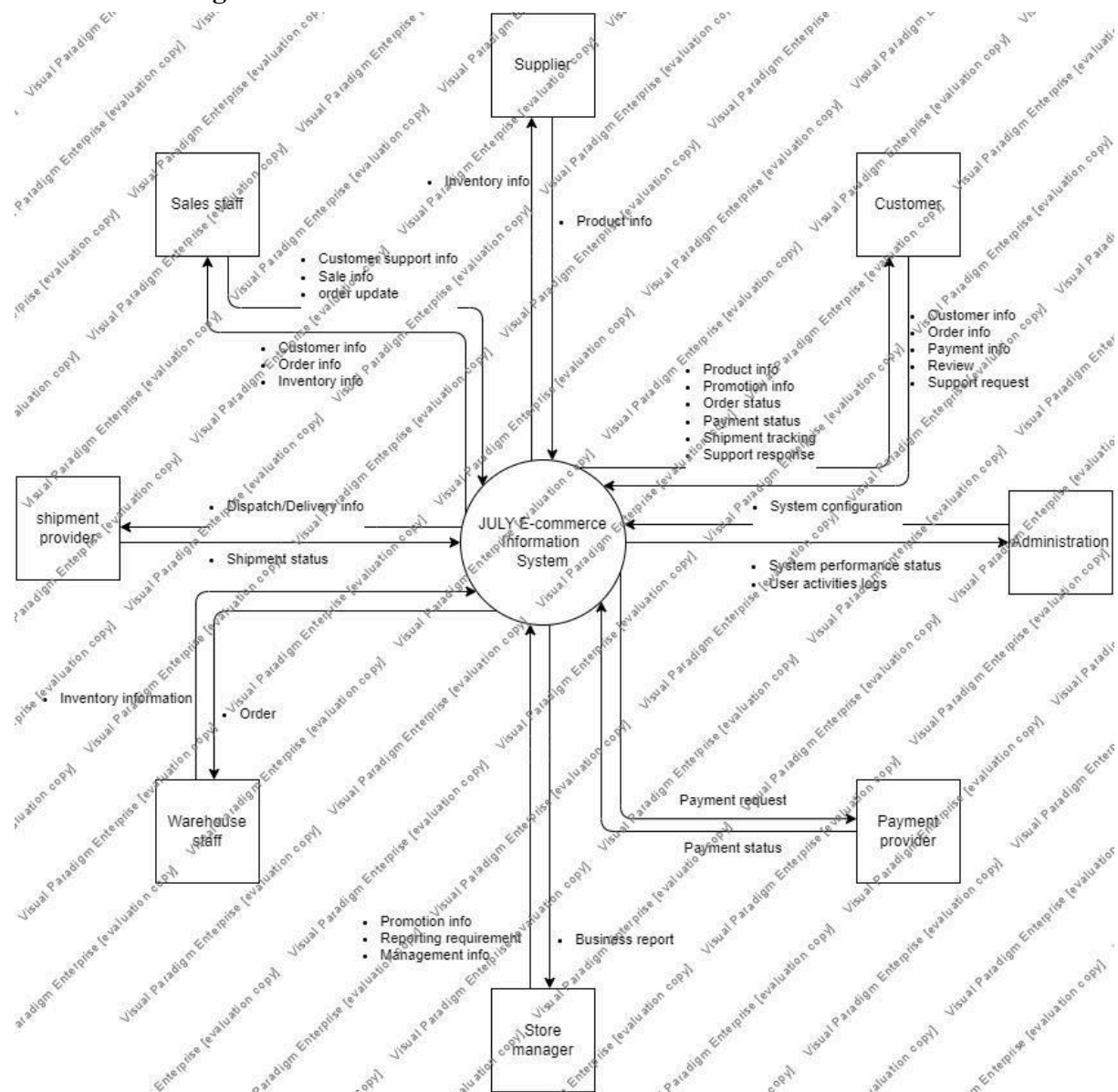


Figure 7.1: Context diagram

7.5. Information Flows

Stakeholder	Direction	Information Exchanged
Customer	→ System	Profile, shopping cart, order, payment, promotion, review, support request
Customer	System →	Product, availability, promotion, order status, payment status, shipment status, review, support response
Administrator	→ System	System configurations, User accounts, Technical updates
Administrator	System →	System performance status, Error Logs, User activity logs
Sales Staff	→ System	Customer support, customer inquiries, order updates, sales information
Sales Staff	System →	Customer, product, order, payment status, inventory, shipment, promotion, customer support, sales, business reports
Warehouse Staff	→ System	Inventory, stock availability, order fulfillment information
Warehouse Staff	System →	Product, order, inventory, order fulfillment information
Store manager	→ System	Promotion information, management information, reporting requirements
Store manager	System →	Business report
Payment Provider	System →	Payment request, confirmed order
Payment Provider	→ System	payment status, transaction information
Shipping Provider	System →	Confirmed order, delivery address, dispatch information
Shipping Provider	→ System	Shipment information and status

Supplier	→ System	Product, product details, supply information
Supplier	System →	Inventory information

7.6. System Objectives

The proposed system aims to:

- **O1:** Centralize business information to ensure consistent and accurate product, customer, order, and inventory data.
- **O2:** Improve order and delivery management by providing clear order, payment, shipment, and tracking information.
- **O3:** Reduce manual work and errors in information management and business reporting.
- **O4:** Improve customer service through unified customer information and support management.
- **O5:** Support management decision-making with timely and reliable business reports.

Current Problem	Related Objective(s)
P1. Product information is inconsistent across the catalogue.	O1
P2. Inventory information is inaccurate or outdated.	O1, O3
P3. Order processing involves excessive manual work.	O2, O3
P4. Order status and tracking information are unclear.	O2
P5. Payment and shipment information is not effectively synchronized.	O2
P6. Customer information is duplicated or fragmented across records.	O1, O4

P7. Customer inquiries and support information are difficult to manage consistently.	O4
P8. Business reports are slow to produce and prone to errors.	O3, O5
P9. Management has limited visibility into sales, orders, inventory, and customer activity.	O5

8. Conclusion

This report identified five key problems at JULY Online Shop:

- Inaccurate or outdated inventory information.
- Inconsistent product information.
- Unclear order status and tracking.
- Fragmented customer information.
- Slow and error-prone reporting.

The proposed E-Commerce Information System will address these problems through five objectives (**O1–O5**): centralizing information, improving order and delivery management, reducing manual work and errors, improving customer support, and supporting management decision-making. These findings provide the foundation for **Assignment 02** and the development of detailed system requirements.

9. References

[1] Tran, D. Q. (n.d.). Information System Analysis and Design - Assignment 01: E-Commerce Information System Investigation. Department of Information Technology, PTIT.